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StatWise Analytics Macular Degeneration

STAT 469

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Agenda

- Introduction
- Exploratory Data Analysis
- Modeling
- Inference
- Prediction
- Conclusion

Presenters

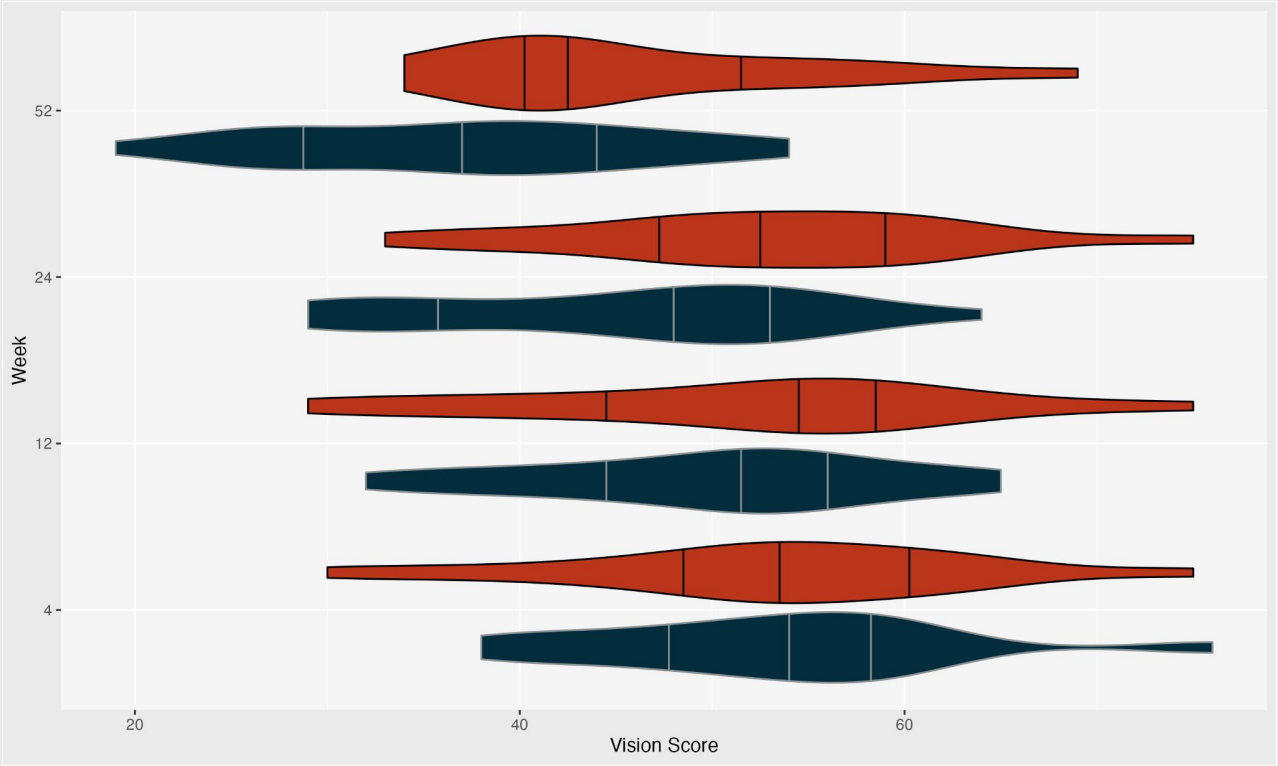
Member			
Role	**Statistician** Exploratory Data Analysis	**Statistician** Modeling	**Statistician** Inference & Prediction

Introduction

Project Objectives

1. Explore trends in vision over time
2. Build an effective longitudinal model
3. Evaluate treatment effect on decline
4. Compare outcomes at 52 weeks

Violin Plot of Vision Scores and Treatments

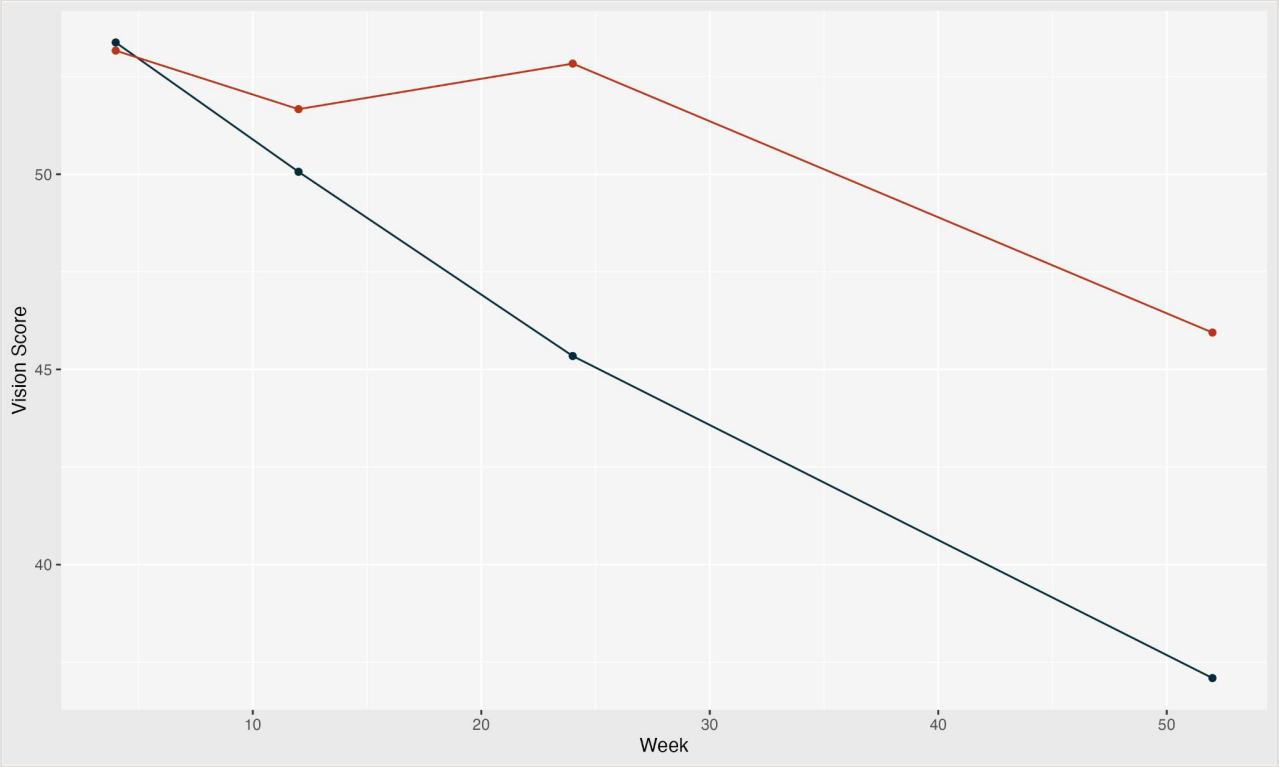


Seeing the Distribution of Vision Score Between Treatment Type

- At week 4, vision scores for both groups are similar and tightly clustered.
- Over time, the placebo group's scores shift downward and the distribution widens, reflecting greater vision loss and variability.
- In contrast, the interferon-α group maintains higher medians and tighter distributions, showing slower degeneration and better preservation of vision by week 52.

Time Series of Vision Scores

Interferon α Placebo



Comparing Vision Scores between Treatment Types

- Both groups start with similar vision scores around 53 letters at week 4.
- Over time, the placebo group steadily loses vision, dropping to 37 letters by week 52, while patients receiving interferon-α maintain higher scores and decline more slowly, ending around 46 letters.
- This divergence suggests that interferon-α helps slow vision loss compared to no treatment.

Possible Modeling Approaches

General Linear Model

$$y \sim MVN(X\beta, \sigma^2 I)$$

- y: dependent variable
- X: predictor variable matrix
- β: effects of predictor variables
- σ^2: variance
- I: Identity matrix

- Doesn't account for between subject correlation
- leading to biased standard errors and unreliable hypothesis tests and confidence intervals.

Longitudinal Model

$$y \sim MVN(X\beta, \sigma^2 \mathbf{B})$$

- All the same as the General Linear model except:
- B instead of I
- What is B?

What is B?

Block Diagonal Matrix With 4×4 Correlation Blocks (R)

$$\begin{bmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} \\ \rho_{31} & \rho_{32} & 1 & \rho_{34} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 \end{bmatrix}$$

Why It's Important

- All the individual ρ's account for the within subject correlation
- Which leads to unbiased errors and reliable hypothesis tests
- This will give us a better idea on how effective interferon-α actually is on preventing vision loss.

$$y \sim MVN(X\beta, \sigma^2 B)$$

X Matrix

- y is the vision score over time
- We included:
  - Time
  - Treatment
  - Original baseline score
  - Time and treatment interaction
  - Time and baseline interaction
  - Treatment and baseline interaction

β Matrix

- The β Matrix includes:
  - intercept
  - The effect of time
  - The effect of treatment
  - The effect of time and treatment interaction
  - The effect of time and baseline interaction
  - The effect of treatment and baseline interaction

Does interferon-α stop vision loss?

Interaction Effect (Time × Treatment)

Estimate: 0.178

95% CI: (0.083, 0.274)

p-value: 0.0002

Rate of Change in Vision Score (Placebo vs Treatment)

Placebo: -0.34/week

Treatment: -0.16/week

Evaluating Treatment Effect on Vision Decline

- The interaction between time and treatment is significant (p = 0.0002).
- Patients on interferon-α still lose vision, but more slowly over time.
- Vision still declines in both groups, but more slowly with treatment.
- Rate of decline for the placebo group is about 0.34 per week versus a rate of decline of 0.16 per week for those with treatment
- This shows the treatment significantly slows vision loss.

Predicted Vision Score at 52 Weeks

Placebo: 37.1

Interferon-α: 45.9

Difference: **+8.8**

Comparing Vision Outcomes at 52 Weeks

- At 52 weeks, the treatment group has higher vision scores.
- Interferon-α: ~45.9
- Placebo: ~37.1
- This is almost an improvement of a score of +9 with treatment.

Conclusion

Project Objectives

1

Trends: Downwards, Normally Distributed

2

Model: Longitudinal over Linear Model

3

Evaluate: Significant Time and Treatment Interaction

4

Predict: Treatment will yield higher score

Thank you!

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If you have any questions, please contact us via email.